WORKED SOLUTION

STEP II 1995, Question 10

Mechanics • Collisions and Restitution

STEP II 1995, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/95-s2-q10

The key idea

Each collision is a two-equation problem: conservation of momentum and Newton's experimental law. Do the two collisions in turn, set the second sphere's final velocity equal to u, and solve for m_1; the inequality then comes from the simple fact that a mass must be positive. For the final part, notice that collisions only care about *relative* velocities, so watching the motion from a frame moving at speed v turns it into the first part.

Solution

We model the spheres as particles moving along one line on a smooth table, so the only horizontal forces are the impulses between spheres. Take the direction of the first sphere's motion as positive.



The first collision

Let the velocities of the first and second spheres just after the first collision be v_1 and w. Momentum and Newton's law (the speed of separation is e times the speed of approach) give

$$m_1 v_1 + m_2 w = m_1 u, \qquad w - v_1 = eu.$$

Substituting $v_1 = w - eu$ into the first equation, $(m_1 + m_2)w = m_1(1 + e)u$, so

$$w = \frac{m_1(1+e)u}{m_1 + m_2}, \qquad v_1 = \frac{(m_1 - em_2)u}{m_1 + m_2}.$$

The second collision

Now the second sphere, moving at w, hits the third, which is at rest. This is the same situation with m_1, m_2, u replaced by m_2, m_3, w, so the second sphere's velocity afterwards is

$$\frac{(m_2 - em_3)w}{m_2 + m_3}.$$

We are told this equals u. Substituting for w and cancelling $u \neq 0$,

$$\frac{(m_2 - em_3)\, m_1(1+e)}{(m_2+m_3)(m_1+m_2)} = 1, \qquad \text{so} \qquad m_1(1+e)(m_2 - em_3) = (m_1+m_2)(m_2+m_3).$$

Collecting the terms in m_1,

$$m_1[(1+e)m_2 - e(1+e)m_3 - m_2 - m_3] = m_2(m_2+m_3),$$

and the bracket simplifies to $em_2 - m_3(1+e+e^2)$. Hence

$$\boxed{m_1 = \frac{m_2(m_2+m_3)}{em_2 - m_3(1+e+e^2)}}.$$

Ask yourself Where does an inequality come from, when all we have is an equation?

From the physics: masses are positive. The numerator $m_2(m_2+m_3)$ is positive, and the bracket we divided by cannot be zero (otherwise the equation before it would read $0 = m_2(m_2+m_3)$). So $m_1 > 0$ forces the denominator to be positive:

$$em_2 - m_3(1+e+e^2) > 0, \qquad \text{that is,} \qquad m_2 e > m_3(1+e+e^2),$$

as required.

It is worth confirming that nothing else happens afterwards, since the final part counts collisions. After the second collision the second sphere moves at u, while the first moves at $v_1 = u - \frac{m_2(1+e)u}{m_1+m_2} < u$, so the first sphere never catches the second again. The third sphere moves away from the second with relative speed $ew > 0$. So there are exactly two collisions.

The final part

Now all three spheres share a common velocity v except the first, which moves at $u > v$. Look at the motion from a frame moving with constant velocity v along the line. In this frame:

- the first sphere moves at $u - v > 0$ towards the second, and the second and third are at rest;
- momentum is still conserved in each collision, because subtracting v from every velocity changes both sides of the momentum equation by the same amount, $(m_a + m_b)v$;

- Newton's law is unchanged, because it involves only differences of velocities.

So in the moving frame we have exactly the situation of the first part, with u replaced by $u - v$, and with the masses satisfying the same relation (which does not involve u at all). By the first part, after two collisions the second sphere moves at $u - v$ relative to the frame. Returning to the table's frame, we add v back on: the velocity of the second sphere after two collisions is

$$\boxed{u}.$$

The condition $u > v$ is what guarantees the first collision happens at all; the hypothesis $v > 0$ is not actually needed for the argument.

Checks

Take $e = \frac{1}{2}$, $m_3 = 1$ and $m_2 = 4$; then $em_2 = 2 > \frac{7}{4} = m_3(1 + e + e^2)$, and the formula gives $m_1 = \frac{4 \cdot 5}{1/4} = 80$. With $u = 7$, the first collision gives the second sphere $w = \frac{80 \cdot \frac{3}{2} \cdot 7}{84} = 10$, and the second collision then gives it $\frac{(4 - \frac{1}{2}) \cdot 10}{5} = 7 = u$, as it should. A computer also ran the collisions exactly (with fractions) for over 10^4 random admissible choices of masses and e, confirming in each case that there are exactly two collisions, that the second sphere ends at u, and that adding a common velocity $v < u$ to the second and third spheres gives the same final velocity u.

Common slips

- Writing Newton's law the wrong way round, e.g. $v_1 - w = eu$; the sphere in front must end up moving at least as fast as the one behind.
- In the final part, recomputing the collisions from scratch instead of using the change of frame, which the question explicitly invites.
- Forgetting to add v back at the end and giving $u - v$ as the answer.

Beyond STEP

The final part is an example of Galilean invariance: the laws of collisions look the same in every frame moving at constant velocity, so you can choose the frame that makes the problem easiest. In university mechanics the favourite choice is the centre-of-mass frame, in which the total momentum is zero and a collision simply reverses and scales the velocities by e. Kleppner and Kolenkow, *An Introduction to Mechanics*, uses this idea extensively in its treatment of collisions.

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